TENDER SUBMISSION

Geonovum Sensor Data Testbed 2026

Implementation Topic #2: Connect Sensors to SensorThings API Server

Submitted by	Geo Insights V.O.F. | KVK 91255430
Lead contact	Mathis van der Voordt | mathis@geo-insights.nl
Co-applicant	Building Changes B.V. | KVK 56692013 Jan-Willem Weststrate (jw.weststrate@buildingchanges.nl), Guus Pieters (g.pieters@buildingchanges.nl)
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Addressed to	Friso Penninga, Director, Geonovum
Topic	Implementation Topic #2: Connect sensors to SensorThings API server
License	CC BY 4.0 / MIT (source code)

1. Implementation Topic

This submission is for Implementation Topic #2: Connect sensors to a central OGC SensorThings API server.

2. Motivation

Climate adaptation in Dutch cities depends on understanding how green-blue infrastructure actually performs in the field, not just in models or policy documents. Geo Insights builds UrbanAdapt, a platform that supports municipalities and fieldlabs across the full lifecycle of climate adaptation planning: from spatial analysis and scenario comparison to real-time monitoring of deployed measures.

The OGC SensorThings API standard directly addresses one of the core bottlenecks in our domain: sensor data from fieldlabs and urban living labs is fragmented, siloed, and inaccessible to the wider ecosystem of planners, policymakers, and researchers who need it. Standardising connectivity through SensorThings API is the interoperability layer that makes fieldlab data actionable at city scale.

We already implement OGC-compliant endpoints in UrbanAdapt's monitoring module. This testbed gives us the opportunity to formalise, validate, and openly document the connection between real-world climate adaptation sensors and a standardised central server, making the approach reproducible for any municipality or fieldlab in the Netherlands.

Our approach is distinctive within this testbed: rather than industrial process sensors or utility infrastructure, we connect environmental monitoring sensors on nature-based green-blue infrastructure in urban living labs. This use case is directly relevant to the Deltaprogramma Ruimtelijke Adaptatie and is underrepresented in current SensorThings API adoption. The knowledge generated is immediately applicable to the dozens of Dutch municipalities executing climate adaptation strategies under national policy obligations.

Building Changes joins as co-applicant. They support organisations and SMEs in the Dutch construction and infrastructure sector through transition processes, operating at the intersection of climate adaptation, circular construction, and digital innovation. Their network in the bouw & infra sector, including via regional knowledge networks, enables dissemination to practitioners and municipalities well beyond the geo-data community.

3. Use Case & Sensor Locations

We will connect sensors from two confirmed living lab locations, with two additional locations in preparation. Each represents a distinct context for green-blue infrastructure monitoring, providing diversity of sensor type, environment, and governance. This maximises learning value for Geonovum.

Our connector is fully compatible with the Brabantse Delta FROST server provided for this testbed. We will push all sensor observations to this central server as our primary endpoint. During Phase 1 we will coordinate with the Topic #1 server contractor to align our SensorThings API data model, ensuring consistent use of Things, Sensors, ObservedProperties, and Datastreams across implementations and enabling cross-use-case querying on the shared server. This maximises the interoperability value of the testbed as a whole.

3.1 The Green Village | TU Delft, Delft (confirmed)

The Green Village is an open innovation fieldlab on the TU Delft campus where experiments with sustainable technologies and nature-based solutions are conducted in real-world conditions. Geo Insights has an active partnership with The Green Village, including a completed pilot project and published research on cost-benefit analysis of climate adaptation measures. Sensor data access has been confirmed by Lindsey Schwidder (VPdelta / The Green Village), who acts as the contact for sensor infrastructure across both TGV and Diergaarde Blijdorp.

Setup / location	Parameters measured	BGI relevance
Hitteplein	Temperature (°C), Humidity (% RH), Solar radiation (W/m²)	Heat stress on urban surface materials
Rainaway Parkeerplaatsen	Soil moisture (m³/m³), Soil temperature (°C) at 6 depths across 2 plots	Infiltration & retention under permeable paving
Flowsand	Soil moisture (m³/m³), Soil temperature (°C)	Permeable substrate water behaviour
Koers / Zoak / Nus	Soil moisture, Soil temperature, Electrical conductivity (mS/cm)	Comparative soil moisture across soil types
Climate Davis (weather station)	Temperature, Precipitation (mm), Wind speed (m/s), Wind direction (°)	Ambient climate baseline
Weather Climatics (rooftop)	Temperature, Precipitation, Air pressure (hPa), Wind speed, Wind direction	Secondary weather reference

3.2 Diergaarde Blijdorp | Rotterdam (confirmed)

Diergaarde Blijdorp has been a VPdelta / TU Delft field lab for climate adaptation and water innovation since 2014. The 28-hectare site in central Rotterdam functions as an urban delta, comparable in scale to a small neighbourhood, and is instrumented for ongoing monitoring of climate adaptation measures including a climate-adaptive forecourt with Bufferblocks, a Bluebloqs water purification system, heat stress monitoring, and water quality monitoring. Lindsey Schwidder (VPdelta) manages the sensor infrastructure at this location and will facilitate data access.

Setup / location	Parameters measured	BGI relevance
Microclimate sensors (animal enclosures)	Temperature (°C), Humidity (% RH)	Urban heat stress in a green, biodiverse urban context
Weather station	Wind speed (m/s), Solar radiation (W/m²)	Site-level climate baseline for Rotterdam
Klimaatplein / Bufferblocks	Water buffer level, infiltration performance	Rainwater capture, storage and reuse monitoring
Water quality (ponds)	Water quality parameters (to be specified with Lindsey Schwidder)	Urban water management and ecological quality

3.3 EUR Campus & Living Lab Binckhaven (in preparation)

A third location at the Erasmus University Rotterdam Campus is in preparation, with a confirmed weather station and soil moisture sensors expected to be operational within three weeks. Lindsey Schwidder is facilitating access to this location. Living Lab Binckhaven (The Hague), an active project, is also in discussion for participation in this testbed. Confirmation is expected within four weeks. Our contact person is Bjorn Jansen. Both are candidates for inclusion during Phase 2 if access to sensors is confirmed before implementation starts.

4. Plan of Approach

4.1 Technical approach

We will build a generic, open-source SensorThings API connector layer that bridges existing fieldlab sensor infrastructure with the central FROST server. The connector operates as a reusable intermediary translation solution rather than a bespoke point-to-point integration, so that any comparable fieldlab or municipality can adapt it. It ingests sensor observations from existing feeds, maps them to the OGC SensorThings API data model (Things, Sensors, ObservedProperties, Datastreams, Observations), and pushes them to the central server, handling protocol translation transparently.

The UrbanAdapt monitoring module, which already implements OGC-compliant endpoints on our own climate adaptation sensors, serves as the integration layer. Sensor data thereby becomes simultaneously available in our platform and queryable on the central testbed server, providing a live public demonstration from day one of operation. The UrbanAdapt monitoring module is available for review on request during the evaluation process.

- Testbed-specific connector (public): github.com/Geo-insights/sensorthings-testbed-connector
- UrbanAdapt monitoring module (OGC-compliant, climate adaptation use case): github.com/Geo-insights/monitoring_module (Access on request)

4.2 Phase 1 | Inventory & mapping (May, weeks 1-2)

- Document sensor infrastructure at TGV and Blijdorp: protocols, data formats, update frequencies
- Map each sensor feed to the SensorThings API data model
- Identify protocol translation requirements per location
- Coordinate with Topic #1 server contractor to align SensorThings API data model across implementations, enabling cross-use-case querying on the central server
- Finalise connector architecture

4.3 Phase 2 | Connector development (May-June, weeks 3-6)

- Build open-source connector as a configurable, modular bridge supporting multiple sensor locations
- Implement observation push from TGV and Blijdorp to the central FROST server
- Integrate with UrbanAdapt monitoring module for simultaneous platform display and FROST server feed
- Publish source code to GitHub (MIT license) throughout development
- Onboard EUR Campus and/or Binckhaven if access is confirmed in time

4.4 Phase 3 | Validation (June, weeks 6-7)

- Validate accuracy: cross-reference pushed observations against source readings per location
- Validate reliability: minimum 14-day continuous operation, logging uptime, latency, error rate
- Test SensorThings API query performance on the central server for climate adaptation use cases
- Document failure modes and edge cases as input for lessons learned

4.5 Phase 4 | Documentation & demonstration (July)

- Publish step-by-step how-to guide: from fieldlab sensor to SensorThings API observation, written for non-specialist organisations
- Contribute written report to Geonovum GitHub repository in markdown format
- Public demonstration: live sensor data from TGV and Blijdorp queryable on the central FROST server, visualised in UrbanAdapt
- Present at all three Geonovum open testbed sessions (May, June, July) and the final public session
- Publish blog post on geo-insights.nl; distribute via Building Changes network to bouw & infra sector and municipal contacts

4.6 Open standards and interoperability

This implementation contributes to an interconnected chain of Dutch and European spatial data standards. The OGC SensorThings API is consistent with INSPIRE Directive requirements for environmental monitoring data and with the Dutch NEN3610 geo-information framework. By connecting climate adaptation sensor data to SensorThings API, we produce observations directly compatible with the Dutch Digital Twin infrastructure (DTaaS) and the emerging European Common Data Space for Smart Cities and Communities. Our connector's generic architecture also supports future alignment with the INSPIRE Environmental Monitoring Facilities data model, which shares conceptual roots with the OGC Observations, Measurements and Sampling (OMS) standard used in Topic #1. Data produced in this testbed is thereby structurally interoperable with the broader European climate adaptation data ecosystem.

4.7 Re-usability after testbed ends

The connector will remain publicly available on GitHub (MIT license) for at least 6 months post-testbed. The live connection from TGV and Blijdorp will remain publicly queryable on the central server for the same period. UrbanAdapt's commercial integration means maintenance is not dependent on testbed funding: the implementation is embedded in an active product used by fieldlabs. The how-to documentation is written explicitly for reuse by municipalities and engineering consultancies, not a specialist geo-data audience.

Nelen & Schuurmans, a partner of Geo Insights and an active user of OGC standards in their own RANA platform, is already collaborating with Geo Insights on Living Lab Binckhaven. Both parties have expressed the intention to integrate the sensor data and connector developed in this testbed into the further development of UrbanAdapt's monitoring module, providing an additional layer of continuity for the implementation beyond the testbed period.

5. Data Governance & Legal Framework

Given the multi-party structure and public publication requirement, we address data governance explicitly, with a view to scalability beyond the testbed.

5.1 Data ownership

Sensor data remains the property of the respective site operator. The Green Village / VPdelta (TU Delft) owns the Delft sensor data; Diergaarde Blijdorp / VPdelta owns the Rotterdam sensor data. Geo Insights does not claim ownership of sensor observations. Participation agreements with each operator will confirm: (a) permission to transmit observations to the Geonovum testbed server; (b) CC BY 4.0 publication agreement; (c) continuation of public access for 6 months post-testbed. Lindsey Schwidder (VPdelta) has confirmed data access in principle, formal agreements to follow before testbed start.

5.2 GDPR

All sensor data measures environmental and infrastructure parameters (soil moisture, temperature, water retention, air quality). No personal data is collected, processed, or published. GDPR obligations are not applicable to this implementation.

5.3 Intellectual property & licensing

The SensorThings API connector will be published under MIT license. Geo Insights retains IP over the broader UrbanAdapt platform but asserts no IP claims over any deliverable contributed to the testbed. All testbed deliverables are published under CC BY 4.0.

5.4 Scalability

The federated ownership model used here (each site retains data ownership, the standard provides interoperability) is directly replicable as SensorThings API moves toward mandatory adoption in the Dutch public sector. We will document recommended data governance arrangements in our how-to guide to make this aspect of adoption explicit for other municipalities and fieldlabs considering similar implementations.

6. In-Kind Investment

The Geonovum budget contribution for Topic #2 is €5,000 excluding VAT. Geo Insights and Building Changes contribute the following additional in-kind:

Activity	Est. hours	Party
Sensor inventory & protocol mapping	12h	Geo Insights
Connector architecture & development	40h	Geo Insights
Validation & testing	16h	Geo Insights
Documentation & how-to guide	12h	Geo Insights
Testbed sessions & public demonstration	8h	Geo Insights
Dissemination & stakeholder outreach	16h	Building Changes
Partner coordination & sensor access	8h	Geo Insights + Building Changes
Total project value (~112h at €100/hr)	**~112h**	**~€11,200**
Geonovum budget contribution		€5,000 excl. VAT
In-kind contribution (consortium)		**~€6,200**

At a rate of €100/hr, the total project value is approximately €11,200. The Geonovum budget contribution of €5,000 (excl. VAT) is applied toward the direct implementation activities. The remaining ~€6,200 is contributed in-kind by Geo Insights and Building Changes. SensorThings API connector development is part of UrbanAdapt's product roadmap regardless of this testbed, making the in-kind contribution substantive and sustained beyond the testbed period.

7. References & Portfolio

7.1 Geo Insights

- The Green Village project page: Geo Insights pilot
- Publication: 'Een kosten-baten plaatje voor klimaatadaptatie' by The Green Village & Geo Insights
- Active project: Living Lab Binckhaven, The Hague, digital twin and IoT monitoring of climate adaptation measures in a dense urban heat island context
- Partner: Nelen & Schuurmans: OGC-compliant platform, active collaborator on Living Lab Binckhaven, and intended partner in the further development of the UrbanAdapt monitoring module.
- Consortium: Fieldlabs@scale, connecting fieldlab experimentation to city-wide climate adaptation policy
- Testbed connector (public, open-source): github.com/Geo-insights/sensorthings-testbed-connector
- UrbanAdapt monitoring module (OGC-compliant, climate adaptation use case): github.com/Geo-insights/monitoring_module (Access on request)
- Website: www.geo-insights.nl

7.2 Building Changes

- Supports public and private organisations and SMEs in the Dutch construction and infrastructure sector through transition processes
- Active in NL-funded innovation consortia including TKI Bouw & Techniek programme Toekomstgerichte wijk, TKI Deltatechnologie programme Wijkgebonden Watersysteem, NWO-funded Ambitions traject, MOOI circular installations for buildings
- Network spans climate adaptation, biobased & circular construction, material passports & certificates, and digitalization in area developments, transitions of public space en future proof living environments in cities
- Develops Toekomstgerichte wijk living labs which focuses on knowledge development and practical solutions for the climate-resilient design of existing Dutch neighborhoods. The focus there is on realizing a sensory measurement framework real time demonstrating quality of physical interventions. This innovation project builds on existing platform solutions and green and/or blue practical applications that are used in various urban projects
- Website: www.buildingchanges.nl and www.toekomstgerichtewijk.nl
- Building Changes is part of the SIH Groep www.sih.nl

7.3 Sensor locations contacts

Lindsey Schwidder, Project Manager Water Innovations, VPdelta / The Green Village, TU Delft. Contact for sensor data access at The Green Village, Diergaarde Blijdorp and the EUR Campus.

Bjorn Jansen, Project Lead, Living Lab Binckhaven, The Hague. Lecturer and change manager at De Haagse Hogeschool (Climate & Management), with hands-on experience in facilitating experiments and collaboration in public green space. Contact for sensor data access at Living Lab Binckhaven.

8. Curriculum Vitae | Key Performers

Mathis van der Voordt | CEO & GIS Specialist, Geo Insights

- BSc Applied Geo-Information Sciences, HAS Green Academy, Den Bosch (2020-2025). Specialisation in climate adaptation, smart cities, data-driven design, and spatial analysis. Programming: Python, JavaScript, HTML/CSS.
- Co-founder & CEO, Geo Insights (2023–present). Leads product development, financial operations, subsidy applications, sales, and stakeholder management for UrbanAdapt, a climate adaptation platform for municipalities and fieldlabs.
- EMVI Specialist & Product Owner Geo, Integraaljagers (Dec 2024-Nov 2025). Coordinated high-scoring tender submissions across disciplines. Detached as Product Owner Geo at DCMR Milieudienst Rijnmond, leading geospatial product development and translating stakeholder needs into scalable data-driven solutions in a complex public-sector environment.
- Project Manager, The Hague Tech (2023-present). Project manager for the Urban Sensing Lab, Hackathon for Good, and OneGov AI Hackathon Series, directly relevant experience in urban sensing infrastructure and open innovation formats.
- LinkedIn: linkedin.com/in/mathisvandervoordt

Iris van Zoggel | CMO & Project Lead, Geo Insights

- BSc Applied Geo-Information Sciences, HAS Green Academy, Den Bosch (2020-2025). Skills: QGIS, ArcGIS Pro, FME, spatial analysis, data collection and interpretation. Minor: Media Psychology, Hogeschool Utrecht (2023).
- Co-founder, Geo Insights (2023-present). Leads marketing, communications, visual design, and project management. Responsible for client-facing deliverables including decks, one-pagers, and stakeholder reporting.
- GIS Analyst, Socotec Geotechnics (2025-present). Analysing geotechnical data, automating GIS workflows, creating spatial analyses and map layouts for clients, and supporting groundwater monitoring fieldwork.
- Internship, Grodan / Rockwool (2022). Software testing in a greenhouse environment, statistical analysis, investigation reporting, and concept design, demonstrating applied research experience in a sensor-heavy context.
- LinkedIn: linkedin.com/in/iris-van-zoggel-4990a022a/

Iust Kuipers | CTO, Geo Insights

- BSc Technology, Policy & Management, TU Delft (2021–2025). Pre-master Econometrics, Vrije Universiteit Amsterdam (2023–2024). Currently MSc Quantitative Finance (Honours Programme), VU Amsterdam (2025–present).
- Co-founder & CTO, Geo Insights (2024–present). Designed and implemented data-driven models using machine learning and GIS to analyse spatial risk factors related to climate exposure and infrastructure resilience. Built and validated analytical pipelines for integrating heterogeneous datasets, including the UrbanAdapt monitoring module with OGC-compliant sensor data endpoints.
- AI Engineer, AI Pilots (Feb–Jul 2025). Designed and implemented AI-driven automation systems, agent-based models integrating reinforcement learning and LLMs, and adaptive frameworks for data-driven decision-making.
- Innovation Engineer, ProRail / TU Delft (Oct 2024–Feb 2025). Worked on innovative solutions to improve rail infrastructure sustainability, leveraging cutting-edge technologies in collaboration with TU Delft researchers.
- Military Working Student, Ministerie van Defensie / Koninklijke Marechaussee (Jul 2025–present). Designed and implemented a centralised data system for sensitive operational datasets; contributed to data-driven research on national resilience and crisis preparedness.
- Programming: Python (data analysis, modelling, pipelines), R, MATLAB. GitHub: github.com/iustkuipers
- LinkedIn: linkedin.com/in/iust-kuipers-50298a221/

Jan-Willem Weststrate | Building Changes

- Innovation & Project manager Building Changes, Delft (2000-present) Various projects: development and guidance TKI programmes Toekomstgerichte wijk, developing knowledge products and services for organisations and SME's. Implementation of Sustainable Civil Engineering in projects and organizations. Guiding the development and change process for Stadswerk072 / Municipality of Alkmaar. Guidance CROW research into the implementation maturity of decentralized authorities.
- Process and Programme Manager Stichting NG Infra (2000-present). Program Manager of the demand-driven (scientific) research program aimed at connecting the knowledge sector to societal practice issues / challenges faced by infrastructure managers. Support of the SIVOON theme center. Support of various organizational and contractual tasks. Operationally responsible for the process design of the Foundation. Program manager of the demand-driven (scientific) research program aimed at connecting the knowledge sector to societal practice issues / challenges of infrastructure managers.
- Team Manager, Project Organisation Maasvlakte 2 (2009–2014). Responsible for scope, cost, and planning integration across disciplines in one of the Netherlands' largest infrastructure projects.
- Erasmus University Rotterdam, Business Administration program, Strategic Management major, Hogeschool Zeeland in Vlissingen, H.T.S. Civil Engineering program, Hydraulic Engineering specialization (1993-2003)
- LinkedIn: linkedin.com/in/jan-willem-weststrate-a214965/

Guus Pieters | Building Changes

- Senior Advisor Circularity, Climate, Sustainability & Innovation, Building Changes, Delft (2021 - present). Guidance of innovation programs with SME entrepreneurs in construction and infrastructure, focused on digitalization, innovation, sustainability, and collaboration with public clients.
- Program Manager National Projects Academy Neerlands Diep (RWS, Prorail, RVB, G4) (2014-2022). Program Manager Digital Transformation & Technologies
- Initiator/Co-founder Platform Omgevingsmanagement (2011-2015). Director of a knowledge platform for environmental managers and agencies that request and provide contributions to this. Developing various activities relevant to the target group and the environment
- Social innovation entrepreneur (2007-2022). Advice and guidance to organizations on transitions, sustainability, and innovation.
- Program Manager Stichting Bouw Research SBR (2000-2007) Development of research programs and knowledge sharing in the construction sector. Environment & Innovation Lobbyist General Association of Construction Companies AVBB (Bouwend Nederland) (1997-2000). Advocacy and policy development in the field of environment and innovation.
- Consultant Heidemij (Arcadis) (1990-1997). Advice on environmental issues and sustainable solutions.
- Consultant AT Kearney (1989-1990). Advice on environmental issues and sustainable solutions.
- Study Environmental Engineering, Wageningen University (1979-1988)
- LinkedIn: linkedin.com/in/guuspieters/

9. License Statement

Geo Insights and Building Changes agree to publish all research results, reports, data, and deliverables produced within this testbed under the Creative Commons Attribution 4.0 International License (CC BY 4.0): http://creativecommons.org/licenses/by/4.0/.

All source code developed for this testbed will be published under the MIT open-source license, as identified by the Open Source Initiative. All deliverables will remain publicly available for a minimum of six months after completion of the testbed.

Signed on behalf of Geo Insights V.O.F. and Building Changes B.V. :

Mathis van der Voordt - CEO, Geo Insights
29 April 2026

Remko Matsinger - Operational Manager, Building Changes
29 April 2026